

Courier Billing Audit & Logistics Cost Analysis Dashboard

Project Overview

The Courier Billing Audit & Logistics Cost Analysis Dashboard is an end-to-end Power BI project designed to analyze courier billing accuracy and logistics costs. The dashboard compares expected courier charges with actual billed charges to identify billing discrepancies, monitor billing performance, and support cost optimization.

This project demonstrates how data analytics can be used to improve logistics operations, reduce unnecessary expenses, and enhance invoice auditing processes.

Business Problem

Courier companies charge shipping fees based on factors such as delivery zones and shipment weight. Incorrect billing can result in overcharges, undercharges, and increased logistics costs.

The objective of this project is to:

- Verify courier billing accuracy.
 - Identify overcharged and undercharged shipments.
 - Analyze billing discrepancies across delivery zones.
 - Measure billing performance using KPIs.
 - Support data-driven logistics decisions.
-

Tools & Technologies Used

- Microsoft Excel
 - Power BI Desktop
 - Power Query
 - DAX (Data Analysis Expressions)
 - Data Visualization
-

Key Performance Indicators (KPIs)

The dashboard includes the following KPI cards:

1. Total Orders Processed

Displays the total number of shipments analyzed.

2. Total Invoice Amount

Shows the overall courier invoice value.

3. Correctly Charged Orders

Represents orders billed accurately according to expected charges.

4. Overcharged Orders

Displays the number of orders billed above expected charges.

5. Undercharged Orders

Shows the number of orders billed below expected charges.

6. Net Difference

Measures the overall difference between expected and billed charges.



Dashboard Visualizations

1. Order Billing Status Distribution

Purpose:

Provides an overview of billing accuracy by displaying the distribution of:

- Correctly Charged Orders
- Overcharged Orders
- Undercharged Orders

Business Value:

Helps stakeholders quickly understand overall billing performance.

2. Orders Distribution by Delivery Zone

Purpose:

Displays shipment volume across delivery zones.

Business Value:

Identifies zones handling the highest number of shipments.

3. Overcharge vs Undercharge by Zone

Purpose:

Compares overcharged and undercharged amounts across different delivery zones.

Business Value:

Highlights operational areas generating the largest billing discrepancies.

4. Top 10 Overcharged Orders

Purpose:

Displays orders with the highest overcharge amounts.

Business Value:

Allows finance and logistics teams to investigate high-risk billing cases and prioritize invoice audits.

5. Billing Accuracy %

Purpose:

Measures the percentage of correctly billed orders against total orders.

Business Value:

Provides a quick performance indicator of courier billing accuracy.



Interactive Filters (Slicers)

The dashboard includes dynamic slicers to improve user interaction and analysis.

Status

Allows filtering by:

- Correctly Charged
- Overcharged
- Undercharged

Delivery Zone as per X

Enables zone-wise analysis of courier billing performance.

Weight Slab as per X (KG)

Allows evaluation of billing accuracy based on shipment weight categories.

Key Insights Generated

The dashboard helps uncover several important business insights:

- Identification of billing discrepancies.
 - Analysis of courier billing accuracy.
 - Zone-wise shipment distribution analysis.
 - Detection of high-value overcharged orders.
 - Evaluation of logistics cost performance.
 - Monitoring of operational efficiency.
-

Business Impact

This solution helps organizations:

- Improve invoice auditing efficiency.
 - Detect overcharged shipments quickly.
 - Reduce logistics and transportation costs.
 - Improve courier partner accountability.
 - Increase billing transparency.
 - Support data-driven decision making.
-

Skills Demonstrated

This project demonstrates proficiency in:

- Data Cleaning
 - Data Transformation
 - Power Query
 - Data Modeling
 - DAX Calculations
 - KPI Development
 - Dashboard Design
 - Business Intelligence
 - Supply Chain Analytics
 - Logistics Cost Optimization
 - Data Visualization
-

Dashboard Summary

KPI Cards

- ✓ Total Orders Processed
- ✓ Total Invoice Amount
- ✓ Correctly Charged Orders
- ✓ Overcharged Orders
- ✓ Undercharged Orders
- ✓ Net Charge Difference

Visuals

- ✓ Order Billing Status Distribution
- ✓ Orders Distribution by Delivery Zone
- ✓ Overcharge vs Undercharge by Zone
- ✓ Top 10 Overcharged Orders
- ✓ Billing Accuracy %

Slicers

- ✓ Status
- ✓ Delivery Zone as per X
- ✓ Weight Slab as per X (KG)

Conclusion

The Courier Billing Audit & Logistics Cost Analysis Dashboard provides a comprehensive solution for validating courier invoices and monitoring logistics costs. By comparing expected and billed charges, the dashboard enables organizations to identify billing discrepancies, improve financial control, and optimize logistics operations through data-driven insights.